Exercises (with Solutions) · Chapter 2: Probability

Exercises with answer options (a)–(d) are **single-choice** questions: exactly one answer is correct.

1. **Problem**
   Two Coin Tosses (Script: Exercise "Two Consecutive Coin Tosses")

   A coin lands heads with probability $P(H) = 0.7$. The coin is tossed twice, and the two tosses are independent. Let H_1 be the event that heads occurs on toss 1 and let H_2 be the event that heads occurs on toss 2.

   What is the probability that heads occurs on toss 1 **or** on toss 2, i.e., $P(H_1 \cup H_2)$?

   (a) 0.49
   (b) 0.42
   (c) 0.51
   (d) 0.91

   Solution
   Since the two tosses are independent,

   $$P(H_1 \cap H_2) = P(H_1)\,P(H_2) = 0.7 \cdot 0.7 = 0.49.$$

   Thus,

   $$\begin{aligned} P(H_1 \cup H_2) &= P(H_1) + P(H_2) - P(H_1 \cap H_2) \\ &= 0.7 + 0.7 - 0.49 \\ &= 0.91. \end{aligned}$$

   (a) False. This is $P(H_1 \cap H_2)$, i.e., the probability of heads on *both* tosses.
   (b) False. This is the probability of heads on *exactly one* of the two tosses.
   (c) False. This is $1 - P(H_1 \cap H_2)$, i.e., the probability of *not* getting heads on both tosses.
   (d) True.

2. **Problem**
   Disjoint Events (Script: Exercise "Independence vs. Disjoint Events")

   A shop's customers pay by card (event A), in cash (event B), or with their mobile phone (event C). Each customer uses exactly one payment method, such that the three events are disjoint: $A \cap B = \emptyset$, $A \cap C = \emptyset$, and $B \cap C = \emptyset$. Moreover,

   $$P(A) = 0.4 \quad \text{and} \quad P(B) = 0.5.$$

   Which of the following statements is true?

   (a) $P(A \cup B) = P(A)\,P(B) = 0.20$.
   (b) A and B are independent, since $P(A \cap B) = P(A)\,P(B) = 0.20$.
   (c) A and B are *not* independent, since $P(A \cap B) = 0 \neq 0.20 = P(A)\,P(B)$.
   (d) $P(A \mid B) = P(A) = 0.4$.

   Solution
   Since A and B are disjoint, $P(A \cap B) = P(\emptyset) = 0$. However,

   $$P(A)\,P(B) = 0.4 \cdot 0.5 = 0.20 > 0.$$

   Thus, $P(A \cap B) \neq P(A)\,P(B)$, and A and B are not independent. Disjoint events with strictly positive probabilities are *never* independent: knowing that a customer pays by card tells us that this customer does *not* pay in cash.

(a) False. For disjoint events, $P(A \cup B) = P(A) + P(B) = 0.9$.

(b) False. Since $A \cap B = \emptyset$, we have $P(A \cap B) = 0 \neq P(A)\,P(B)$.

(c) True.

(d) False. $P(A \mid B) = P(A \cap B)/P(B) = 0/0.5 = 0$: a customer who pays in cash never pays by card.

3. **Problem**
Independent Events? (Script: Exercise "Deriving Independence")

A fair die is tossed once. Let

$$A = \{2, 4, 6\}$$

be the event of observing an even number and let

$$B = \{1, 2\}.$$

Which of the following statements is true?

(a) $P(A \cup B) = P(A) + P(B)$.

(b) A and B are disjoint, i.e., $A \cap B = \emptyset$.

(c) A and B are independent.

(d) A and B are not independent.

Solution
We check the definition of independence. We have $A \cap B = \{2\}$ and, thus,

$$P(A \cap B) = \frac{1}{6},$$

$$P(A)\,P(B) = \frac{3}{6} \cdot \frac{2}{6} = \frac{1}{6}.$$

Thus, $P(A \cap B)$ equals $P(A)\,P(B)$, and the events A and B are independent.

(a) False. This addition rule only holds for disjoint events. Here, $P(A \cup B) = P(A) + P(B) - P(A \cap B)$ with $P(A \cap B) = \frac{1}{6} > 0$.

(b) False. $A \cap B = \{2\} \neq \emptyset$.

(c) True.

(d) False; see above.

4. **Problem**
Medical Test (Script: Example "Medical Tests")

A disease D occurs in 1% of the population, i.e., $P(D) = 0.01$. A medical test for this disease has outcomes $+$ and $-$, with

$$\text{sensitivity:} \quad P(+ \mid D) = 0.95,$$

$$\text{specificity:} \quad P(- \mid D^c) = 0.9.$$

A person gets a positive test result. What is the probability that this person has the disease, i.e., $P(D \mid +)$? (Answers are rounded to three decimals.)

(a) 0.088

(b) 0.900

(c) 0.109

(d) 0.950

Solution

Step 1: Probabilities of the complements. We need two probabilities that are not given directly, but that follow from the rule $P(A^c) = 1 - P(A)$ for the complement A^c ("not A") of an event A.

- Every person is either sick (D) or healthy (D^c). Thus, the probability of being healthy is

$$\begin{aligned} P(D^c) &= 1 - P(D) \\ &= 1 - 0.01 \\ &= 0.99. \end{aligned}$$

- Every healthy person gets either a negative ($-$) or a positive ($+$) test result. Thus, among the healthy people, the probabilities of these two results add up to one, $P(- \mid D^c) + P(+ \mid D^c) = 1$. (Recall that for fixed D^c, $P(\cdot \mid D^c)$ is a probability.) Therefore, the probability that a healthy person gets a (false) positive test result is

$$\begin{aligned} P(+ \mid D^c) &= 1 - P(- \mid D^c) \\ &= 1 - 0.9 \\ &= 0.10. \end{aligned}$$

Step 2: Joint probabilities. Rearranging the definition of conditional probability, $P(A \mid B) = P(A \cap B)/P(B)$, gives $P(A \cap B) = P(A \mid B)\, P(B)$. Thus, the probability of being sick *and* testing positive is

$$\begin{aligned} P(+ \cap D) &= P(+ \mid D)\, P(D) \\ &= 0.95 \cdot 0.01 \\ &= 0.0095, \end{aligned}$$

and the probability of being healthy *and* testing positive is

$$\begin{aligned} P(+ \cap D^c) &= P(+ \mid D^c)\, P(D^c) \\ &= 0.10 \cdot 0.99 \\ &= 0.099. \end{aligned}$$

Step 3: Probability of a positive test result. A person with a positive test result is either sick or healthy. These two events are disjoint, so we can add their probabilities:

$$\begin{aligned} P(+) &= P(+ \cap D) + P(+ \cap D^c) \\ &= 0.0095 + 0.099 \\ &= 0.1085. \end{aligned}$$

Step 4: Conditional probability. By the definition of conditional probability,

$$\begin{aligned} P(D \mid +) &= \frac{P(+ \cap D)}{P(+)} \\ &= \frac{0.0095}{0.1085} \\ &\approx 0.088. \end{aligned}$$

Intuition: Out of $10{,}000$ people, 100 are sick, and 95 of them test positive. Out of the 9900 healthy people, 990 also test positive. Thus, only 95 of the 1085 positive test results belong to sick people.

(a) True.

(b) False. This is the specificity $P(- \mid D^c)$.

(c) False. This is the probability $P(+)$ of a positive test result.

(d) False. This is the sensitivity $P(+ \mid D)$. Confusing $P(+ \mid D)$ with $P(D \mid +)$ is the prosecutor's fallacy.

5. **Problem**
Drawing Cards (Script: Exercise "Drawing Cards Without Replacement")

Draw two cards from a standard deck of 52 cards *without* replacement. The deck contains 4 aces. What is the probability that **both** cards are aces, i.e., that the first card is an ace **and** the second card is an ace? (Answers are rounded to four decimals.)

(a) $\frac{4}{52} \cdot \frac{3}{51} \approx 0.0045$

(b) $\frac{4}{52} \cdot \frac{4}{51} \approx 0.006$

(c) $\frac{3}{51} \approx 0.0588$

(d) $\frac{4}{52} \cdot \frac{4}{52} \approx 0.0059$

Solution
Let A be the event that the first card is an ace, and let B be the event that the second card is an ace. Then,

$$P(A) = \frac{4}{52}.$$

Given A, the first card is not replaced, so 51 cards remain, 3 of which are aces:

$$P(B \mid A) = \frac{3}{51}.$$

Rearranging the definition of conditional probability gives

$$P(A \cap B) = P(A)\,P(B \mid A) = \frac{4}{52} \cdot \frac{3}{51} \approx 0.0045.$$

(a) True.

(b) False. After the first draw, only 3 aces are left.

(c) False. This is only the conditional probability $P(B \mid A)$.

(d) False. This would be the answer for drawing *with* replacement (independent draws).

6. **Problem**
pmf, cdf, Mean, and Variance (Script: Examples "cdf of a Discrete Random Variable (Car Sales)" and "pmf of a Discrete Random Variable (Car Sales)")

Another car dealership sells $X \in \{0, 1, 2, 3\}$ cars per day with the following probability mass function (pmf):

x	0	1	2	3
$f_X(x) = P(X = x)$	0.4	0.3	0.2	0.1

(a) Check that f_X is a valid pmf.

(b) Write down the cdf $F_X(x) = P(X \leq x)$ for all $x \in \mathbb{R}$ and sketch it.

(c) Compute the probability that the dealership sells at most one car on a given day, and the probability that it sells more than two cars.

(d) Compute the mean $E(X)$, i.e., the average number of cars sold per day.

(e) Compute the variance $\mathrm{Var}(X)$.

Solution

(a) All pmf values are non-negative, and they sum to one:

$$0.4 + 0.3 + 0.2 + 0.1 = 1.$$

(b) The cdf adds up the pmf values of all values smaller than or equal to x:

$$F_X(x) = \begin{cases} 0 & \text{for } x < 0 \\ 0.4 & \text{for } 0 \leq x < 1 \\ 0.7 & \text{for } 1 \leq x < 2 \\ 0.9 & \text{for } 2 \leq x < 3 \\ 1 & \text{for } x \geq 3. \end{cases}$$

The cdf is a step function. It jumps at $x = 0, 1, 2, 3$, and the jump heights are the pmf values $f_X(x)$.

(c) **At most one car:** "At most one car" means $X \leq 1$, i.e., $X = 0$ or $X = 1$. This is exactly the value of the cdf at $x = 1$:

$$\begin{aligned} P(X \leq 1) &= F_X(1) \\ &= P(X = 0) + P(X = 1) \\ &= 0.4 + 0.3 \\ &= 0.7. \end{aligned}$$

More than two cars: "More than two cars" means $X > 2$. The cdf only gives us probabilities of the form $P(X \leq x)$, so we use the complement: the event "more than two cars" ($X > 2$) is the complement of the event "at most two cars" ($X \leq 2$). Every day falls into exactly one of these two events, so their probabilities add up to one:

$$P(X > 2) + P(X \leq 2) = 1.$$

Solving for $P(X > 2)$ gives

$$\begin{aligned} P(X > 2) &= 1 - P(X \leq 2) \\ &= 1 - F_X(2) \\ &= 1 - 0.9 \\ &= 0.1. \end{aligned}$$

Check: More than two cars means exactly three cars here, and indeed $P(X = 3) = 0.1$.

(d) For a discrete random variable, the mean is the probability-weighted sum of its values:

$$\begin{aligned} E(X) &= 0 \cdot 0.4 + 1 \cdot 0.3 + 2 \cdot 0.2 + 3 \cdot 0.1 \\ &= 1.0. \end{aligned}$$

That is, the dealership sells 1.0 cars per day on average.

(e) The variance is the probability-weighted sum of the squared deviations from the mean $E(X) = 1.0$:

$$\begin{aligned}\mathrm{Var}(X) &= (0-1.0)^2 \cdot 0.4 + (1-1.0)^2 \cdot 0.3 \\ &\quad + (2-1.0)^2 \cdot 0.2 + (3-1.0)^2 \cdot 0.1 \\ &= 1.00.\end{aligned}$$

7. **Problem**
Waiting Times (Script: Examples "cdf of a Continuous Random Variable (Customer Arrivals)" and "pdf of a Continuous Random Variable (Customer Arrivals)")
Let $X \in [0, \infty)$ be the waiting time (in minutes) until the next customer calls a company's service hotline. The waiting time is exponentially distributed with an average waiting time of 10 minutes, i.e., its cdf is

$$F_X(x) = \begin{cases} 0 & \text{for } x < 0 \\ 1 - e^{-x/10} & \text{for } x \geq 0. \end{cases}$$

(a) Compute the probability that the next customer calls within 5 minutes.
(b) Compute the probability that the hotline has to wait more than 30 minutes for the next call.
(c) Compute the probability that the waiting time is between 5 and 30 minutes.
(d) Derive the density function f_X of X.
(e) Show that the probability in (a) equals the area under the density f_X between 0 and 5.

Solution

(a) "Within 5 minutes" means $X \leq 5$. This is exactly the value of the cdf at $x = 5$:

$$\begin{aligned}P(X \leq 5) &= F_X(5) \\ &= 1 - e^{-5/10} \\ &= 1 - e^{-0.5} \\ &\approx 1 - 0.61 \\ &= 0.39.\end{aligned}$$

(b) "More than 30 minutes" means $X > 30$. The cdf only gives us probabilities of the form $P(X \leq x)$, so we use the complement: the event "more than 30 minutes" ($X > 30$) is the complement of the event "at most 30 minutes" ($X \leq 30$). Every waiting time falls into exactly one of these two events, so their probabilities add up to one:

$$P(X > 30) + P(X \leq 30) = 1.$$

Solving for $P(X > 30)$ gives

$$\begin{aligned}P(X > 30) &= 1 - P(X \leq 30) \\ &= 1 - F_X(30) \\ &= 1 - \left(1 - e^{-30/10}\right) \\ &= 1 - 1 + e^{-3} \\ &= e^{-3} \\ &\approx 0.05.\end{aligned}$$

(c) The event "at most 30 minutes" ($X \leq 30$) splits into two disjoint parts: "at most 5 minutes" ($X \leq 5$) and "between 5 and 30 minutes" ($5 < X \leq 30$). Thus,

$$P(X \leq 30) = P(X \leq 5) + P(5 < X \leq 30).$$

Solving for $P(5 < X \leq 30)$ gives

$$\begin{aligned} P(5 < X \leq 30) &= P(X \leq 30) - P(X \leq 5) \\ &= F_X(30) - F_X(5) \\ &= \left(1 - e^{-3}\right) - \left(1 - e^{-0.5}\right) \\ &= 1 - e^{-3} - 1 + e^{-0.5} \\ &= e^{-0.5} - e^{-3} \\ &\approx 0.61 - 0.05 \\ &= 0.56. \end{aligned}$$

(d) The density is the derivative of the cdf. For $x > 0$,

$$f_X(x) = F_X'(x) = \frac{d}{dx}\left(1 - e^{-x/10}\right) = \frac{1}{10}\, e^{-x/10},$$

and $f_X(x) = 0$ for $x < 0$.

(e)

$$\begin{aligned} \int_0^5 \frac{1}{10}\, e^{-x/10}\, dx &= \Big[- e^{-x/10} \Big]_0^5 \\ &= -e^{-5/10} - \left(-e^0\right) \\ &= 1 - e^{-0.5} \approx 0.39, \end{aligned}$$

which equals the probability from (a).

8. **Problem**
Joint pmf, Conditional Means, and Independence (Script: Examples "Two-Dimensional pmf (Car Sales and Staffing)", "Law of Total Expectation (Car Sales and Staffing)", and "Checking Independence in Our Examples")

Let $X_1 \in \{0, 1, 2, 3\}$ be the number of cars sold per day by a car dealership, and let $X_2 \in \{1, 2\}$ be the number of sales consultants on duty that day. The joint pmf $f(x_1, x_2) = P(X_1 = x_1, X_2 = x_2)$ is given by the following table:

$f(x_1, x_2)$	$x_1 = 0$	$x_1 = 1$	$x_1 = 2$	$x_1 = 3$
$x_2 = 1$	0.20	0.20	0.10	0.00
$x_2 = 2$	0.05	0.15	0.20	0.10

(a) Compute the marginal pmf f_1 of X_1 and the marginal pmf f_2 of X_2.

(b) Compute the conditional means $E(X_1 \mid X_2 = 1)$ and $E(X_1 \mid X_2 = 2)$. Interpret the result.

(c) Compute the unconditional mean $E(X_1)$ in two ways: (i) using the marginal pmf f_1, and (ii) using the Law of Total Expectation.

(d) Are X_1 and X_2 independent? Justify your answer.

Solution

(a) The marginal pmfs are obtained by summing the joint pmf over the other variable, i.e., by summing the columns (for f_1) and the rows (for f_2) of the table:

x_1	0	1	2	3
$f_1(x_1)$	0.25	0.35	0.30	0.10

x_2	1	2
$f_2(x_2)$	0.50	0.50

For instance, $f_1(0) = f(0,1) + f(0,2) = 0.20 + 0.05 = 0.25$.

(b) The conditional means are computed with the conditional pmf $f(x_1 \mid x_2) = f(x_1, x_2)/f_2(x_2)$:

$$E(X_1 \mid X_2 = 1) = \sum_{x_1} x_1 \frac{f(x_1, 1)}{f_2(1)}$$
$$= \frac{0 \cdot 0.20 + 1 \cdot 0.20 + 2 \cdot 0.10 + 3 \cdot 0.00}{0.50} = \frac{0.40}{0.50} = 0.800,$$

$$E(X_1 \mid X_2 = 2) = \sum_{x_1} x_1 \frac{f(x_1, 2)}{f_2(2)}$$
$$= \frac{0 \cdot 0.05 + 1 \cdot 0.15 + 2 \cdot 0.20 + 3 \cdot 0.10}{0.50} = \frac{0.85}{0.50} = 1.700.$$

On days with one consultant, the dealership sells 0.800 cars on average; on days with two consultants, it sells 1.700 cars on average.

(c) (i) Using the marginal pmf f_1:

$$E(X_1) = 0 \cdot 0.25 + 1 \cdot 0.35 + 2 \cdot 0.30 + 3 \cdot 0.10 = 1.25.$$

(ii) Using the Law of Total Expectation, we average the conditional means, weighted by the probabilities of X_2:

$$\begin{aligned} E(X_1) &= E\big(E(X_1 \mid X_2)\big) \\ &= E(X_1 \mid X_2 = 1)\, f_2(1) + E(X_1 \mid X_2 = 2)\, f_2(2) \\ &= 0.800 \cdot 0.50 + 1.700 \cdot 0.50 \\ &= 1.25. \end{aligned}$$

(d) No. Independence requires $f(x_1, x_2) = f_1(x_1)\, f_2(x_2)$ for *all* pairs (x_1, x_2). This fails, for instance, for $(x_1, x_2) = (0, 1)$:

$$f(0,1) = 0.20 \neq 0.125 = 0.25 \cdot 0.50 = f_1(0)\, f_2(1).$$

This is also visible in (b): the conditional mean of X_1 depends on X_2, which is impossible if X_1 and X_2 are independent.

9. **Problem**
Two-Dimensional Density (Script: Example "Two-Dimensional pdf (Customer Arrivals and Service Times)")

A company runs a service hotline. Let

- $X_1 \geq 0$ be the waiting time (in minutes) until the next call, and
- $X_2 \geq 0$ be the duration (in minutes) of this call.

The random vector $X = (X_1, X_2)^\top$ has the joint density function

$$f(x_1, x_2) = \begin{cases} \frac{1}{6}\, e^{-x_1/6} \cdot \frac{1}{3}\, e^{-x_2/3} & \text{for } x_1 \geq 0 \text{ and } x_2 \geq 0 \\ 0 & \text{otherwise.} \end{cases}$$

(a) Show that f integrates to one.

(b) Compute the probability that the next call comes within 6 minutes **and** lasts at most 6 minutes.

(c) Compute the marginal density $f_1(x_1)$ of the waiting time X_1.

(d) Are X_1 and X_2 independent? Justify your answer.

Solution

(a) Since the joint density is a product of a function of x_1 and a function of x_2, the double integral factorizes:

$$\begin{aligned} \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f(x_1, x_2)\, dx_1\, dx_2 &= \left(\int_0^{\infty} \frac{1}{6}\, e^{-x_1/6}\, dx_1 \right) \cdot \left(\int_0^{\infty} \frac{1}{3}\, e^{-x_2/3}\, dx_2 \right) \\ &= \Big[-e^{-x_1/6} \Big]_0^{\infty} \cdot \Big[-e^{-x_2/3} \Big]_0^{\infty} \\ &= (0+1) \cdot (0+1) = 1. \end{aligned}$$

(b) Probabilities are volumes under the density surface:

$$\begin{aligned} P(X_1 \leq 6, X_2 \leq 6) &= \int_0^6 \int_0^6 \frac{1}{6}\, e^{-x_1/6} \cdot \frac{1}{3}\, e^{-x_2/3}\, dx_1\, dx_2 \\ &= \left(\int_0^6 \frac{1}{6}\, e^{-x_1/6}\, dx_1 \right) \cdot \left(\int_0^6 \frac{1}{3}\, e^{-x_2/3}\, dx_2 \right) \\ &= \Big[-e^{-x_1/6} \Big]_0^{6} \cdot \Big[-e^{-x_2/3} \Big]_0^{6} \\ &= \left(-e^{-6/6} - \left(-e^{0} \right) \right) \cdot \left(-e^{-6/3} - \left(-e^{0} \right) \right) \\ &= \left(1 - e^{-1} \right) \cdot \left(1 - e^{-2} \right) \\ &\approx 0.63 \cdot 0.86 \approx 0.55. \end{aligned}$$

(c) We integrate out x_2. For $x_1 \geq 0$,

$$\begin{aligned} f_1(x_1) &= \int_{-\infty}^{\infty} f(x_1, x_2)\, dx_2 \\ &= \frac{1}{6}\, e^{-x_1/6} \cdot \underbrace{\int_0^{\infty} \frac{1}{3}\, e^{-x_2/3}\, dx_2}_{=1} \\ &= \frac{1}{6}\, e^{-x_1/6}, \end{aligned}$$

and $f_1(x_1) = 0$ for $x_1 < 0$. Likewise, $f_2(x_2) = \frac{1}{3}\, e^{-x_2/3}$ for $x_2 \geq 0$ and $f_2(x_2) = 0$ otherwise.

(d) Yes. The joint density equals the product of the marginal densities,

$$f(x_1, x_2) = f_1(x_1) \cdot f_2(x_2) \quad \text{for all} \quad x_1, x_2 \in \mathbb{R}.$$

Thus, X_1 and X_2 are independent: the waiting time until the next call tells us nothing about the duration of this call.

10. **Problem**
Regression Function (Script: Example "Conditional Mean Function of a Bivariate Normal Distribution")

The revenue growth X_1 of a firm and the GDP growth X_2 (both in %) are jointly normally distributed with

$$\mu = \begin{pmatrix} E(X_1) \\ E(X_2) \end{pmatrix} = \begin{pmatrix} 2 \\ 0.5 \end{pmatrix} \quad \text{and} \quad \Sigma = \begin{pmatrix} \mathrm{Var}(X_1) & \mathrm{Cov}(X_1, X_2) \\ \mathrm{Cov}(X_1, X_2) & \mathrm{Var}(X_2) \end{pmatrix} = \begin{pmatrix} 1 & 0.4 \\ 0.4 & 2 \end{pmatrix}.$$

For the bivariate normal distribution, the regression function is

$$m(x_2) = E(X_1 \mid X_2 = x_2) = E(X_1) + \frac{\mathrm{Cov}(X_1, X_2)}{\mathrm{Var}(X_2)} \big(x_2 - E(X_2)\big).$$

Which of the following is the regression function $m(x_2)$?

(a) $m(x_2) = 1.90 + 0.20\, x_2$
(b) $m(x_2) = 0.10 + 0.20\, x_2$
(c) $m(x_2) = 1.80 + 0.40\, x_2$
(d) $m(x_2) = 1.86 + 0.28\, x_2$

Solution
Plugging in the parameters gives

$$\begin{aligned} m(x_2) &= 2 + \frac{0.4}{2}\,(x_2 - 0.5) \\ &= 2 + 0.20\,(x_2 - 0.5) \\ &= 1.90 + 0.20\, x_2. \end{aligned}$$

Interpretation: Each additional percentage point of GDP growth increases the expected revenue growth of the firm by 0.20 percentage points.

(a) True.
(b) False. Here, the roles of $E(X_1)$ and $E(X_2)$ were interchanged when computing the intercept.
(c) False. Here, the covariance was divided by $\mathrm{Var}(X_1)$ instead of $\mathrm{Var}(X_2)$.
(d) False. Here, the slope is the correlation $\mathrm{Corr}(X_1, X_2) = 0.28$ instead of $\mathrm{Cov}(X_1, X_2)/\,\mathrm{Var}(X_2)$.

11. **Problem**
Covariance, Regression Function, and the Law of Total Expectation (Script: Examples "Covariance and Correlation of a Bivariate Normal Distribution", "Conditional Mean Function of a Bivariate Normal Distribution", and "Law of Total Expectation (Bivariate Normal Distribution)")

A car manufacturer's revenues react strongly to the business cycle. Let X_1 be the manufacturer's annual revenue growth and let X_2 be the annual GDP growth (both in %). Suppose that

$$X = \begin{pmatrix} X_1 \\ X_2 \end{pmatrix} \sim \mathcal{N}_2\,(\mu, \Sigma) \quad \text{with} \quad \mu = \begin{pmatrix} 3 \\ 1 \end{pmatrix} \quad \text{and} \quad \Sigma = \begin{pmatrix} 4 & 1.2 \\ 1.2 & 1 \end{pmatrix}.$$

(a) What are the standard deviations of X_1 and X_2? Compute the correlation $\mathrm{Corr}(X_1, X_2)$.

(b) Compute the regression function $m(x_2) = E(X_1 \mid X_2 = x_2)$ using

$$m(x_2) = E(X_1) + \frac{\mathrm{Cov}(X_1, X_2)}{\mathrm{Var}(X_2)} \big(x_2 - E(X_2)\big).$$

(c) Interpret the slope of the regression function. What is the expected revenue growth in a year with zero GDP growth, and in a year with 3 percent GDP growth?

(d) Verify the Law of Total Expectation, $E\big(E(X_1 \mid X_2)\big) = E(X_1)$, for this example.

Solution

(a) The standard deviations are $\sqrt{\mathrm{Var}(X_1)} = \sqrt{4} = 2$ and $\sqrt{\mathrm{Var}(X_2)} = \sqrt{1} = 1$ percentage points. The correlation is

$$\mathrm{Corr}(X_1, X_2) = \frac{\mathrm{Cov}(X_1, X_2)}{\sqrt{\mathrm{Var}(X_1)}\sqrt{\mathrm{Var}(X_2)}} = \frac{1.2}{2 \cdot 1} = 0.6.$$

This is a strong positive correlation.

(b)

$$\begin{aligned} m(x_2) &= 3 + \frac{1.2}{1}\,(x_2 - 1) \\ &= 1.8 + 1.2\,x_2. \end{aligned}$$

(c) Each additional percentage point of GDP growth increases the expected revenue growth of the car manufacturer by 1.2 percentage points. Since the slope is larger than one, the manufacturer's revenues react more than one-to-one to the business cycle. Moreover,

$$m(0) = 1.8 + 1.2 \cdot 0 = 1.8,$$

$$m(3) = 1.8 + 1.2 \cdot 3 = 5.4.$$

That is, the expected revenue growth is 1.8 percent in a year with zero GDP growth, and 5.4 percent in a year with 3 percent GDP growth.

(d) If we do not fix the value of X_2, the conditional mean is the random variable $E(X_1 \mid X_2) = m(X_2) = 1.8 + 1.2\,X_2$. Using the linearity of the mean, $E(a + b\,X_2) = a + b\,E(X_2)$,

$$\begin{aligned} E\big(E(X_1 \mid X_2)\big) &= E\big(1.8 + 1.2\,X_2\big) \\ &= 1.8 + 1.2\,E(X_2) \\ &= 1.8 + 1.2 \cdot 1 \\ &= 3 = E(X_1). \end{aligned}$$

12. **Problem**

Mean Independence vs. Independence (Script: Section "Independence Implies Mean Independence (but Not Vice Versa)")

Let X_1 be the daily return of a stock (in %), and let $X_2 \in \{1, 2\}$ describe the state of the stock market on that day: calm ($X_2 = 1$) or volatile ($X_2 = 2$), where

$$P(X_2 = 1) = 0.8 \quad \text{and} \quad P(X_2 = 2) = 0.2.$$

Given the state of the market, the return is normally distributed:

$$X_1 \mid X_2 = 1 \sim \mathcal{N}\,(0.2,\, 1)\,,$$

$$X_1 \mid X_2 = 2 \sim \mathcal{N}\,(0.2,\, 4)\,.$$

(a) Compute the conditional means $E(X_1 \mid X_2 = 1)$ and $E(X_1 \mid X_2 = 2)$.

(b) Compute the unconditional mean $E(X_1)$ using the Law of Total Expectation.

(c) Is X_1 mean independent of X_2?

(d) Compute the conditional variances $\mathrm{Var}(X_1 \mid X_2 = 1)$ and $\mathrm{Var}(X_1 \mid X_2 = 2)$. Are X_1 and X_2 independent? Interpret the result.

Solution

(a) The conditional means are the means of the conditional normal distributions:

$$E(X_1 \mid X_2 = 1) = 0.2 \quad \text{and} \quad E(X_1 \mid X_2 = 2) = 0.2.$$

(b) By the Law of Total Expectation,

$$\begin{aligned} E(X_1) &= E\big(E(X_1 \mid X_2)\big) \\ &= E(X_1 \mid X_2 = 1)\, P(X_2 = 1) + E(X_1 \mid X_2 = 2)\, P(X_2 = 2) \\ &= 0.2 \cdot 0.8 + 0.2 \cdot 0.2 \\ &= 0.2. \end{aligned}$$

(c) Yes. Since

$$E(X_1 \mid X_2 = x_2) = 0.2 = E(X_1) \quad \text{for all } x_2 \in \{1, 2\},$$

X_1 is mean independent of X_2: the state of the market does not affect the *expected* return.

(d) The conditional variances are the variances of the conditional normal distributions:

$$\mathrm{Var}(X_1 \mid X_2 = 1) = 1 \neq 4 = \mathrm{Var}(X_1 \mid X_2 = 2).$$

Thus, the conditional distribution of X_1 given $X_2 = x_2$ depends on x_2, and X_1 and X_2 are *not* independent. Knowing that the market is volatile does not change the expected return, but it makes large gains and large losses more likely. This illustrates that mean independence does not imply independence.